

Concurrent Disseminated Cryptococcus and CMV Chorioretinitis in a Patient with AIDS: A Multidisciplinary Approach to a Complex Case

Vasu Malhotra, DO¹, Shreya Patel, DO¹; Corey Thurman, MD¹; Jessica Cobain, PharmD¹; Imad Dibo, MD¹

¹Lakeland Regional Health, Lakeland, Florida, USA

Introduction

Cryptococcal meningitis and cytomegalovirus (CMV) chorioretinitis are well-recognized opportunistic infections in patients with AIDS; however, concurrent presentation poses significant diagnostic and therapeutic challenges. This case highlights the importance of multidisciplinary collaboration in addressing both acute infectious concerns and long-term healthcare access for a critically ill, immunocompromised patient.

Case Presentation

A 30-year-old male with AIDS (CD4 count = 6) and a history of nonadherence to highly active antiretroviral therapy (HAART) presented with severe headache, persistent vomiting, and altered mental status. Physical exam revealed confusion, neck stiffness, and multiple new skin disseminated lesions. Laboratory evaluation revealed leukopenia (WBC 2.3 K/uL), anemia (Hgb 10.3 g/dL), and thrombocytopenia (112 K/uL). A lumbar puncture revealed a positive cryptococcal antigen titer of 1:1280, with CSF cultures growing *Cryptococcus neoformans*. Opening pressures measured >64 cm H₂O, requiring aggressive ICP management with serial lumbar punctures. Given the severity of his symptoms, the patient was treated with an extended induction phase of IV liposomal amphotericin B and high dose flucytosine. The decision to extend induction therapy was guided by the patient's persistently elevated intracranial pressures, profound immunosuppression, and significant cryptococcal burden as evidenced by persistent fungemia. Serial lumbar punctures successfully lowered ICP, and ongoing aggressive hydration with IV fluids mitigated amphotericin-induced nephrotoxicity. Electrolyte monitoring revealed progressive hypokalemia and hypomagnesemia, necessitating frequent supplementation. During hospitalization, the patient reported vision changes, and ophthalmologic evaluation revealed bilateral chorioretinitis concerning for CMV despite a low CMV DNA PCR (<200 IU/mL). Empiric IV ganciclovir was initiated. The decision to proceed with treatment despite low viremia reflects the understanding that CMV chorioretinitis can occur with minimal or undetectable systemic viral loads.

Discussion

This case underscores the value of multidisciplinary collaboration in managing complex infectious diseases. Coordination between infectious disease specialists, pharmacists, case management, and the health department was essential not only for acute treatment but also for ensuring successful outpatient follow-up. The pharmacist played a key role in medication reconciliation and dosing adjustments, particularly in response to Amphotericin B-induced nephrotoxicity. Aggressive electrolyte repletion and close renal monitoring was crucial to prevent any further sequelae stemming from medication therapy. Case Management and the health department facilitated a seamless transition to outpatient care, securing a follow-up appointment with the Florida Department of Health to improve access to HAART. Extensive counseling with the patient was undertaken to ensure understanding of the importance of medication adherence in preventing future opportunistic infections.

Conclusion

This case highlights several key points in the management of immunocompromised patients. First, low CMV viral loads do not exclude CMV chorioretinitis, and clinical judgment should guide empiric treatment in high-risk patients. Second, extending the induction phase of cryptococcal meningitis treatment may be necessary in severe cases, particularly in patients with persistently elevated ICP. Balancing antifungal efficacy with strategies to mitigate Amphotericin B toxicity is crucial, as nephrotoxicity and electrolyte imbalances can develop despite proactive measures. Finally, addressing social determinants of health is vital in patients with barriers to care, and collaboration with support services plays a key role in achieving long-term clinical stability.

References

